SEDS 520 Microservice Project Report Format

Each team must prepare **report** for its assigned microservice or component.

1. Service Information

This section must include:

- Microservice / Component name
- Team members
- Main responsibility
- GitHub repository link

The GitHub repository link is **mandatory**. The repository must include the source code, README file, and running instructions.

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2. Service Description

Briefly describe the service/component.

This section should answer:

- What does this service/component do?
- What is its main role in the system?
- Which other services does it interact with?

For the Queue team, this section should explain the role of the messaging infrastructure in the overall system.

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3. Requirements / User Stories

List the main functions implemented by the service.

ID	Requirement / User Story
R-01	The service shall register buyers and cooperative members.
R-02	The service shall publish an event after successful registration.

Example for the Queue team:

ID	Requirement
R-01	The queue component shall enable asynchronous communication between microservices.
R-02	The queue component shall route auction-related events to the relevant services.

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4. API / Interface Design

Describe the APIs or interfaces provided by the service.

For regular microservices:

Method	Endpoint	Purpose
POST	/users/register	Registers a new user
GET	/users/{id}	Gets user information

For the Queue team, instead of API endpoints, the following can be described:

Interface	Purpose
Exchange / Topic	Routes events between services
Queue	Stores messages for consuming services
Routing Key	Determines which service receives which event

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5. Data Model / Message Model

Regular microservice teams should describe their data model.

Entity	Main Fields
User	userId, name, email, role
Product	productId, species, quantity, quality

The Queue team should describe the message/event model.

Event Name	Main Fields
BuyerRegistered	eventId, buyerId, name, email
AuctionStarted	eventId, auctionId, startTime
BasketSold	eventId, basketId, buyerId, finalPrice

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6. Events and Communication

Describe the events produced and consumed by the service.

Event	Produced / Consumed	Related Service	Purpose
BuyerRegistered	Produced	Notification Service	Sends registration notification
AuctionStarted	Consumed	Auction Service	Starts the related notification process

The Queue team should especially explain:

- The queue/message broker structure used
- Queue/topic/exchange names
- Which events are routed to which services
- The message flow between services

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7. Implementation Summary and Team Contribution

Provide a brief technical summary and team member contributions.

Include:

- Programming language / framework
- Database, if used
- Queue/message broker, if used
- Main implemented features
- GitHub repository structure

Team Member	Contribution
John	Registration API and database design
Anna	Login functionality and README preparation

At the end of this section, briefly summarize the current status of the service, completed parts, and any missing parts.

Submission Rules

- Report format: PDF
- GitHub repository: **mandatory**
- README file: **mandatory**
- The README must explain how to run the service/component
- Each team submits only the report for its own microservice/component.